Alvara Protocol

[Problem/Solution & Background White Paper — release 1 —

June 18th, 2022 - The Alvara Team]

Abstract

This paper presents the Alvara Protocol. Alvara is a cryptocurrency ETF creation and trading Protocol, built around a hybridized exchange, the HiveX. The protocol decentralizes traditional ETF processes through the establishment of a decentralized network of crypto-based ETFs (SmartFunds) and fund managers. NFT integration and built-in fee sharing structures incentivize fund managers to design, develop and deploy their own crypto SmartFunds, competing against other decentralized fund managers on Alvara's leaderboard. With an NFT "key" attached to each fund, managers can build their fund, earn through the fee share and then sell the management rights and revenue stream via the Alvara NFT marketplace. The ecosystem also provides reward mechanisms for early adopters via staking and yield farming platforms.

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1. Overview

1.1 Introduction

‘Alvara’ / al-VUH-ra / - derived from the Latin “alvarium”, meaning ‘beehive’; also Gothic, meaning ‘guard of all’.

Alvara is a cryptocurrency exchange-traded fund (ETF) creation and trading protocol that utilizes blockchain technology to decentralize traditional ETF processes. The protocol consists of three separate entities:

1. The HiveX (the exchange)
2. The Alvara Hive (the fund)
3. The Alvara Hive Management Company (the fund manager)

Alvara’s vision is to accelerate the decentralization of the global economy by delivering the world’s most popular financial product - the ETF - to the cryptocurrency market.

The traditional financial systems provide a diverse range of investment vehicles. Before the rise of blockchain technology, these OTC services had been provided primarily by centralized financial institutions. One of the most widely offered of these investment vehicles are ETFs. As of the time of writing, ETFs have not made any significant inroads into blockchain or decentralized finance.

1.1.2 Exchange-Traded Funds

An ETF is a form of investment fund not dissimilar to a mutual fund. ETFs track an index or a basket of underlying assets, and as the name infers, their shares are tradable on exchanges, just like stocks. Unlike mutual funds, ETFs are tradable at any time between market open and close or, with respect to the cryptocurrency markets, 24 hours a day, 7 days a week.

ETFs offer distinct advantages over their mutual fund cousins including more flexible trading, greater transparency and better tax efficiency.[11] They also offer advantages when compared to investing in single assets, including **cost efficiency, tradability, diversification, liquidity, and again, tax efficiency**. Commonly seen ETF structures include: index ETFs, actively managed ETFs, inverse ETFs, leveraged ETFs and thematic ETFs.

Over the past 2 decades, ETFs have become an increasingly popular choice for investors. The typical ETF investor in the US tends to be younger, more highly educated and earn a higher income than investors of other financial products.[2] Globally, 50% of investors' portfolios now hold some exposure to exchange-traded funds.

Unlike mutual fund shares, ETF shares are traded on an exchange and are purchased from, and sold to, other investors. This contrasts with mutual funds, whereby investors buy, sell and redeem their shares directly from the fund.

In traditional finance, when an ETF has an initial public offering, or wants to issue additional shares, it does so by selling shares to one or more financial institutions known as Authorized Participants (APs). APs are typically large broker-dealers and are the only parties permitted to create and redeem shares of the ETF. They can do so only in large aggregations or blocks commonly called "creation units" or a "creation basket". Each ETF sets their own specifications for how big a creation basket is.

To purchase shares from an ETF, an AP assembles and deposits a designated basket of securities and cash with the fund in exchange for which it receives shares in the ETF. Once the ETF shares are received, the authorized participant is free to sell the ETF shares in the secondary market to individual investors, institutions, or market makers.

The redemption process is the reverse of the creation process. An AP buys a large block of ETF shares on the open market and delivers those shares to the fund. In return, the AP

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[1] (Carrel & Ferri, n.d.)
[2] (//, n.d. ici.org)

receives a pre-defined basket of individual securities, or the cash equivalent. Most ETFs charge the authorized participants a fee for each of these "creation basket" transactions, but it is usually nominal.

To summarize, the benefits of ETF investing include:

- lower fees
- instant diversification
- liquidity
- tax efficiency
- sector investing
- the ability to purchase in small amounts
- the availability of a wide variety of alternative investments

1.1.3 ETF History and Mechanics

ETFs were born of the mutual fund industry, the origins of which date back to the late 1700s. The first open-ended fund that provided the mechanisms of creation and redemption of fund shares was the Massachusetts Investment Trust (MIT) in 1924.[3]

The oldest-surviving ETF that trades today is the SPDR S&P 500 ETF (SPY) which tracks the total market index. Other popular ETFs include the NASDAQ-100 index, Nippon Gold ETF, the Russell 1000 Index, and the Invesco QQQ ETF.

As of 2017, there were 5,024 ETFs trading globally, 1,756 based in the United States with over half of the inflows going to the 20 largest ETFs. As of February 2022, assets under management by US ETFs totalled over $5.4 trillion, and globally over $10 trillion, marking the ETF as the world's number one investment vehicle, surpassing its mutual fund cousin for first place.

<u>Transparency</u>

[3] (Davidson, 2012)

With the exception of certain actively managed ETFs, a fund's underlying assets, as well as their weightings, are posted daily on the website of the issuer.

Fees

The largest ETFs have annual fees of 0.03% AUM or lower, although specialty ETFs can have annual fees as high as 1%. These fees are generally paid to the ETF issuer, taken from dividends received from the underlying holdings or from the sale of underlying assets.

Structure

An ETF divides ownership into shares held by shareholders. The shareholders indirectly own the assets of the fund, and they will typically receive annual reports. ETF holders are entitled to a share of the profits, such as interest or dividends, and would be entitled to any residual value if the fund goes into liquidation.

1.1.4 ETFs Today

At the end of 2021, exchange-traded fund assets worldwide reached a value of $10 trillion,[4] with the three largest ETF industries valued at:

- North America: $5.6 trillion
- Europe: $1.3 trillion
- Asia Pacific: $0.69 trillion

As of February 2022, the SPDR S&P 500 ETF Trust is the highest valued ETF globally with a market capitalization of approximately $397 billion.

The Bank of America projects that the Global ETF industry will hold a value of $50 trillion by 2030.[5] This does not take into account the rise of the digital asset industry.

[4] (*ETFs - Statistics & Facts*, n.d.)
[5] (Eckett, 2019)

Digital assets by comparison make up only a small fraction of the global economy. Cryptocurrency's total market cap is a mere 4% of the global financial services markets, and only 10% of the market cap of the global ETF industry. At the beginning of 2022, the total market capitalization of all cryptocurrencies was approximately $2.4 trillion. Presently, there are approximately 13,000 cryptocurrencies in circulation with a total combined market cap of just under $1 trillion.

1.1.5 Defining the Challenges

Alvara is breeding the next generation of digital asset managers, redesigning the crypto investment markets, and solving several of the major problems prevalent in the cryptocurrency landscape:

High Risk Sector:

Cryptocurrency's volatility discourages new investors. **Alvara SmartFunds help manage this volatility through providing vehicles for diversified investment, thereby decreasing investors' exposure to individual high-risk crypto assets.**

Spoilt for Choice:

Investors entering crypto are overwhelmed by a huge range of options - *"It's everything you don't understand about money combined with everything you don't understand about computers,"* - John Oliver.[6] **Alvara solves this problem with its network of experienced and verifiable fund managers. Users can check the performance history of each manager and make a decision based on their profitability. This simplifies investment decisions for less experienced traders, or those who simply do not have the time to follow a multitude of assets. Alvara strives to help traders invest their time, and capital, wisely.**

Slow Adoption:

The vast majority of global assets under management (AUM) remains in centralized TradFi markets. Offering the world's most popular investment vehicle, **Alvara attracts investment**

[6] (Dillet, 2018)

from traditional markets, facilitating the mass adoption of cryptocurrency.

Lack of Verification:

A major problem in the crypto space is the havoc wreaked by a small community of disreputable trader-influencers. These "guns-for-hire" will openly support any paying project, regardless of its credentials, often resulting in poor investment decisions by their followers. There are also many trustworthy, genuine and skilled trader-KOLs (key opinion leaders) with strong performance history who will only support and recommend bona fide projects. Unfortunately without an easy means for verifying the genuine from the scandalous, all KOLs tend to be tarred with the same brush, at least from the perspective of the wider community and investors from the traditional markets. This lack of verification and accountability cultivates a poor and untrusting sentiment within the landscape, and is arguably a major inhibitor to mass adoption. **Alvara provides a platform where the genuine traders-come-influencers can clearly and precisely demonstrate their skills, and platform users can verify a fund manager's performance history directly within the protocol, leaving no ambiguity as to the validity of their claims. Alvara will separate the skilled from the unskilled in an exciting, gamified setting, breeding a new generation of influencers, and giving support to the many legitimate trader-influencers in the space.**

1.2 ALVA Token

ALVA is the protocol's native token. It is an ERC-20 utility token that serves multiple functions within the platform, from fee reductions to portal access. ALVA is also included in each SmartFund listed on the platform at a nominal weighting. This has a deflationary effect on the token supply, as with each additional SmartFund that is created, more ALVA is removed from the circulating supply. The included ALVA is market-bought, which increases buying pressure with each SmartFund creation.

ALVA has a maximum supply of 200 million. The original ERC-20 version of the token can be both minted and burned via the smart contract. The full ALVA distribution, capital raise and emission schedules can be found here.

1.2.1 Token Utility

The ALVA token has five significant use cases within the Alvara Protocol.

1. Fee reductions:

All platform fees paid using the ALVA token will attract a 25% fee reduction.

2. Portal access:

Design portal: Users must hold a minimum of 5,000 ALVA in order to access the design portal in v2. Without this minimum holding requirement, they will not be able to design and list their own SmartFund.

Whitelist portal: Users wishing to take part in the initial creation of each SmartFund must hold a minimum balance of 5,000 ALVA in their wallet. The advantages to taking part in the initial creation are a guaranteed allocation of the SmartFund at the NAV price, with initial creation fees shared between all participants.

3. Staking rewards:

The Alvara staking platform is discussed in detail in section 2.3.2

4. Farming rewards:

Alvara farming initiatives are discussed in detail in section 2.3.3

5. Inclusion in every SmartFund:

The ALVA token will make up a maximum of 5% of every SmartFund listed on the platform. The exact percentage will depend on the size of the SmartFund and the ALVA liquidity pool, but 5% will be the standardized weighting. Included ALVA will always be market-bought, putting upwards pressure on the market price while simultaneously locking up a percentage of the circulating supply.

1.2.2 Advanced Tokenomics

The Alvara Treasury:

The treasury will hold a reserve of ALVA tokens and stable coins that will be used as required to cover expenses of the protocol, once the capital raised during the presale rounds has been exhausted. The treasury comprises 30% of the total supply, locked over a 24 month period from TGE. The treasury will also receive 80% of all platform transaction fees, taken from the trading of Alvara SmartFunds, and 55% of all fees taken from the trading of 3rd party SmartFunds.

ALVA holders:

ALVA holders can earn a passive income simply by holding the ALVA token. They can also generate additional yield via Alvara's staking platform, and partner yield farming platforms. A sum of ALVA tokens, equivalent to 10% of all transaction fees, will be gift-airdropped to ALVA holders on a monthly basis, weighted according to their ALVA holdings. I.e. if a user holds 1% of the total ALVA supply, they will receive 0.1% equivalent of the HiveX transaction fees (1% of the 10% distributed to ALVA holders).

The Alvara Foundation:

The Alvara Foundation will be a UK registered charity, established to fight for our world's wild bee and pollinating insects populations. The charity will be registered prior to the launch of the HiveX platform, and 5% of all platform fees will be paid into its treasury. If paid in ALVA, the balance will then be sold on the open market on a monthly basis (randomized dates to avoid market manipulation) to fund the Foundation's endeavors. This ALVA sale is offset by an equally weighted burn event, detailed below. The Alvara Foundation's initial plans will be to partner with and provide financial support to other established organizations working to help support bee populations around the world.

ALVA Burn Program:

Alvara's burn program is designed to decrease the total ALVA token supply over time, increasing the value of each ALVA token. 5% of all platform fees taken in the form of ALVA will be burned on a monthly basis, offsetting the sale of ALVA going into the Alvara Foundation.

ALVA BuyBack & Burn Program:

5% of all fees taken that are paid using stable coins will be used to purchase ALVA on the open market on a monthly basis (randomized dates to prevent market manipulation). These ALVA tokens will then be burned, reducing the total ALVA supply.

ALVA as an underlying asset:

Every SmartFund listed on Alvara's HiveX will contain the ALVA token at a weighting of 5%. This ALVA will be purchased on the open market, rather than being taken from a reserve. This feature will put upwards pressure on ALVA's market price each time a new SmartFund is launched, as well as each time additional SmartFund shares are created by platform users. It also removes a percentage of the ALVA token's supply from the circulating market.

Platform fees will be distributed as follows:

For Alvara SmartFunds:

- *80% paid into the Alvara treasury*
- *10% equivalent gift-airdropped to ALVA holders*
- *5% paid into the Alvara Foundation*
- *5% burned via the Alvara Burn Program if paid in ALVA. If paid in stable coin, the remaining 5% will be used in the Alvara BuyBack & Burn Program*

For 3rd party SmartFunds:

- *55% paid into the Alvara treasury*
- *25% paid to the SmartFund's manager*
- *10% equivalent gift-airdropped to ALVA holders*
- *5% paid into the Alvara Foundation*
- *5% burned via the Alvara Burn Program if paid in ALVA. If paid in stable coin, the remaining 5% will be used in the Alvara BuyBack & Burn Program*

1.2.3 Token Distribution

The ALVA token has been distributed proportionally within the ecosystem as follows:

Allocation	Percentage (%)	No. ALVA
Seed Round	**12%**	**24,000,000 ALVA**
Series A Private Round	10%	20,000,000 ALVA
Series B Public Round	5%	10,000,000 ALVA
Operational	30%	60,000,000 ALVA
Liquidity	10%	20,000,000 ALVA
Team	12%	24,000,000 ALVA
Advisors	5%	10,000,000 ALVA
Community Rewards	2%	4,000,000 ALVA
Staking Rewards	7%	14,000,000 ALVA
Farming Rewards	7%	14,000,000 ALVA
Total	**100%**	**200,000,000 ALVA**

2. Ecosystem

At the core of the Alvara Protocol is the HiveX - a hybridized, centralized exchange - designed specifically for the creation, redemption and trading of cryptocurrency

exchange-traded funds. Being a hybrid exchange, the HiveX leverages both blockchain and off-chain technology to deliver secure high-speed and low-cost trading.

In addition to the HiveX, the ecosystem consists of an **ALVA staking platform**, designed for early adopters to generate yield by staking their ALVA tokens on the platform; a **farming rewards initiative**, provided by a third party to incentivize ALVA holders to contribute to the DEX liquidity pools; **The Alvara Hive**, a social trading platform granting users the option to connect and track one another's portfolios, building an interconnected structure of influential portfolio managers and investors, and; the **Alvara Lab**, providing educational content and tools that will allow users to experiment and develop their fund management skill set in a synthesized setting.

2.1 Blockchains

2.1.1 Ethereum

As the undisputed home of decentralized finance, Ethereum will be home to the ALVA token and protocol. ALVA will initially be launched as an ERC-20 token on the Ethereum blockchain.

2.1.2 Avalanche

Alvara will also employ the use of the Avalanche blockchain to launch a mirrored version of the ALVA token. The official Avalanche Bridge (AB) (https://bridge.avax.network/) will be the bridge of choice for executing this process. The team will make an application for ALVA to be listed on the bridge after launch.

Avalanche C-Chain's EVM (Ethereum Virtual Machine) supports the deployment of all

Ethereum smart contracts, affording users all the benefits of building on Ethereum, but with the advantage of Avalanche's sub-second transaction speeds (measured by time-to-finality), and nominal transaction fees.

Avalanche is utilized for the HiveX's on-chain features, including deposits and withdrawals, the creation and redemption mechanisms, NFT integration and DeFi yield generating activities.

2.2 HiveX

Alvara's hybrid cryptocurrency exchange, the HiveX, facilitates the trading, creation and management of crypto SmartFunds. The protocol allows users to trade Alvara SmartFunds as well as to design and manage their own. Fund creators are rewarded for their SmartFund designs and management via lucrative fee-sharing structures.

The more successful and popular a SmartFund, the higher the trading volumes, and the larger the gross trading and management fee take. All fund managers are ranked on a leaderboard which provides investment credibility and a performance-validated track record.

Powered by its Golang matching engine (GME), the HiveX can process over 500,000 transactions per second (TPS). This, coupled with on-chain integration provides competitive, rapid and low cost transactions to users whilst maintaining the security and transparency that blockchain provides.

V1: The HiveX V1 will host Alvara-built and managed SmartFunds.

V2: The HiveX V2 will host all V1 SmartFunds and will introduce user-built and managed SmartFunds.

2.2.1 Hybridization

Alvara's HiveX is a hybrid exchange (HEX), meaning that it employs both onchain and offchain technology. By combining all of the advantages of centralized exchanges (speed, cost, security) with all of the advantages of blockchain technology (immutability, transparency, trust), hybrid exchanges offer an enhanced trading experience to users.

2.3 Reward Distribution

2.3.1 Community Reward Supply

Four million ALVA (2% of the total supply) has been allocated to rewarding the community - funding trading competitions, social engagement initiatives, airdrops and marketing campaigns.

2.3.2 Staking

Alvara's staking platform is a single token staking initiative that is designed to reward early holders. ALVA token holders can access the platform directly via the website and stake their tokens securely in exchange for 12.5% APR. Rewards accumulate every block with no locking or cliff period for either claiming rewards or withdrawing initial capital.

Functions include: staking, unstaking, claiming rewards, and compounding rewards. Fees: fees will be applied for each function as follows:

Function Fee (%)
Staking 1.5%
Unstaking 0.5%
Claiming Rewards 0.5%
Compounding Rewards 0.5%

50% of the fees collected by the staking platform are fed back into the operational token supply; with the remaining 50% added to the DEX liquidity pools on a monthly basis.

2.3.3 Farming

Alvara will employ the use of third party farming platform providers to incentivize ALVA holders to contribute liquidity to the DEX pools. In exchange for contributing a liquidity position and then staking those LP tokens on the platforms, users will earn rewards in the form of the ALVA token.

2.4 The Alvara Hive

The Alvara Hive is the protocol's social trading platform. Features include a built in messaging service, chat rooms, the ability to display trading performance history, funds under management and profitability as a fund manager.

The hive will develop a sense of community, belonging and collaboration between users, providing an incentive to keep coming back that is not purely monetary.

2.5 The Alvara Lab

The Alvara Lab is an educational center that aims to de-risk users' investment choices and fund compositions by explaining the inner workings and complexities of exchange-traded funds. The center lets users experiment with the Alvara Analysis Tool (AAT), our proprietary backtesting software tool that is used during the design and development of Alvara SmartFunds. more information on the AAT can be found in section 3.8.

3. Alvara Core Architecture

3.1 Introduction

This section discusses the features and components of Alvara's core architecture - the HiveX. It describes summary detail for each feature of the exchange and Protocol. It is intended to give a brief overview and explanation of how the product's different parts work together. It does not go into full technical detail as that is covered in Alvara's System Requirements Specification (SRS) document.

3.2 HiveX Features

3.2.1 Trading

Trading will enable users to buy and sell Alvara's SmartFund shares: SmartFund/USDT, or other supported stable coin pairs. Traders are able to buy and sell ALVA/USDT directly on the exchange.

Users will have the ability to place a range of different orders:

- Limit Buy
- Limit Sell
- Market Buy
- Market Sell
- OCO: An OCO order consists of a stop-limit order and a limit order with the same order quantity. Both orders must be either buy or sell. If a trader cancels one of the orders, the entire OCO order pair is canceled.
- Stop-Limit

Users can cancel trades at any time providing they have not been executed.

Trading fees: 0.1% on every trade, charged in USDT at time of trade execution and sent to the Alvara platform fee wallet. Users have the option to pay trading fees (and all platform fees) in the ALVA token, reducing fees by 25%.

NAV (Net Asset Value) is also displayed here to provide information regarding whether the SmartFund is trading at a premium or a discount to the NAV.

3.2.2 Deposit Cryptocurrency

Users can deposit ALVA and supported stable coins into their HiveX wallet. Supported stable coins can be used to buy SmartFund shares on the exchange, or to create additional SmartFund shares via the creation mechanism. ALVA or supported stable coins are used to pay the various fees within the HiveX platform. For more information on platform fees please refer to section 3.4.

3.2.3 Withdraw Cryptocurrency

Verified users can withdraw their supported stable coins or ALVA tokens from their HiveX wallet, back to an on-chain wallet or other CEX wallet.

3.2.4 Creation Mechanism

The creation mechanism allows users to create (mint) new SmartFund shares at the NAV of the underlying assets.

The function serves as an upwards pressure release valve. I.e. if a SmartFund is trading at a premium to the NAV, a participant can create additional SmartFund shares at a

discount to the market price and then list these shares on the exchange higher than NAV but lower than trading price to bring the market price back towards equilibrium with the NAV as selling pressure for the SmartFund increases.

Fees: User pays all fees associated (gas/exchange/transfer).

User inputs investment amount. Alvara algorithms calculate the net quantity of SmartFund shares the user can create with input investment.

3.2.5 Redemption Mechanism

The redemption mechanism allows users to redeem (burn) SmartFund shares in exchange for the value of the underlying assets, transferred to the user in ALVA, USDT or other supported stable coins.

The function serves as a downwards pressure release valve. I.e. if a SmartFund is trading at a discount to the NAV, a participant can redeem SmartFund shares at a premium to the market price. This incentive will increase buying pressure for the SmartFund shares and help bring the market price back into equilibrium with the NAV.

Fees: User pays all fees associated (gas/exchange/transfer).

User inputs SmartFund share redemption quantity. Alvara algorithms calculate the net USDT value a user can redeem.

3.3 SmartFunds

Alvara SmartFunds are the name for our crypto ETFs. Offering investors the world's most popular investment vehicle, but on the blockchain. ETFs are number one for a reason - they provide a low risk,

low cost, diversified and transparent investment opportunity. Alvara SmartFunds have several distinct advantages when compared to traditional ETFs:

Build your own

Design and launch your own Crypto SmartFund via Alvara's creation portal. Access Alvara's proprietary backtesting analysis software to help design your fund.

Yield Generation

Every smart fund contains 100% collateralized underlying assets. These assets are utilized in yield generation strategies via our partners, with the rewards being fed back into the funds and distributed to shareholders as dividends.

Proof of Ownership

NFT integration provides smart fund managers with a key to their fund's management rights and rewards structure. Transferable at Alvara's NFT marketplace, fund managers can, for the first time, sell their business.

Advanced Leaderboard

Compete against your fellow fund managers for top ranking positions on Alvara's leaderboard. Follow the most profitable fund managers to minimize your risk exposure to the crypto markets.

Low Cost

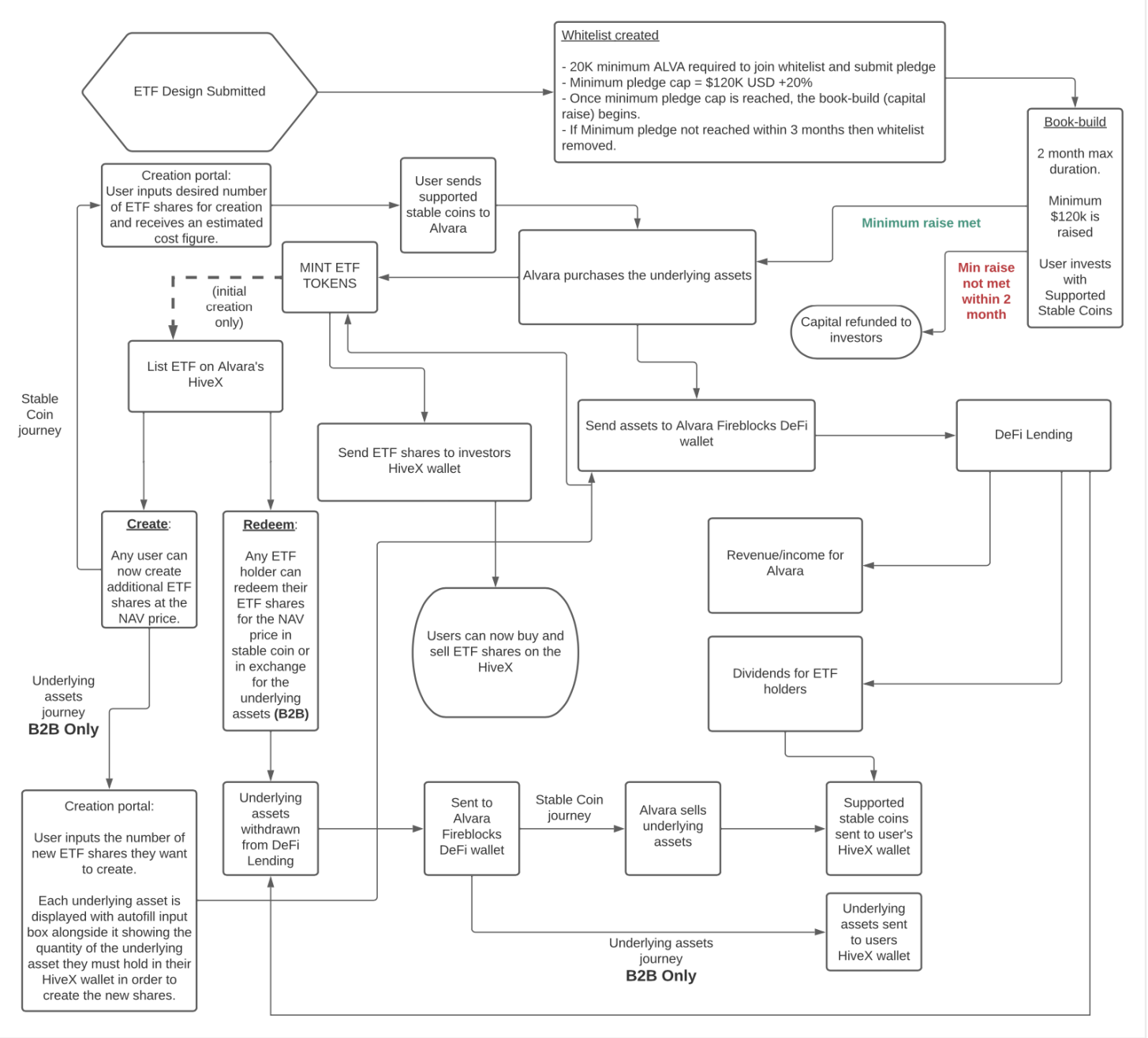
Underlying assets are purchased through APIs linked to leading CEXs, making the costs of creating, redeeming and rebalancing much cheaper than fully on-chain alternatives.

3.4 Business Model

Alvara's business model is based on income derived from charging users platform fees. There are standard transaction fees for trading on the exchange; fees for withdrawing cryptocurrencies out of the exchange; fees for the creation and redemption of SmartFund shares, and; a management fee taken on a percentage of assets under management (AUM).

Fee Type Fee (%)
Transaction 0.1%
Withdrawal 0.1%
Creation/Redemption 0.5%
Management (AUM) 0.5%

3.5 SmartFund Inception to Listing:



3.6 NAV and Calculation

An Alvara SmartFund's NAV represents the net value of the fund and is calculated as the

total value of the SmartFund's assets minus the total value of its liabilities, divided by the number of SmartFund shares.

The NAV of each SmartFund will be used to determine whether or not a SmartFund is trading at a premium or a discount. If market price > NAV then the SmartFund is trading at a premium; if market price < NAV then the SmartFund is trading at a discount.

Correct determination of the NAV is very important as it is the calculation drawn upon when users are utilizing the creation and redemption features.

$$^{7}NAV = \frac{Total\ Value\ Underlying\ Assets - SmartFund\ Liabilities}{Total\ Number\ of\ Outstanding\ SmartFund\ Shares}$$

3.7 Yield Generation

The underlying assets of all Spot SmartFunds within the platform will be available to employ in yield generation strategies.

The revenue generated will first be used to cover any other operating and maintenance costs not already covered by the fund and the remainder will be distributed to the SmartFund holders as dividends on a monthly basis.

Alvara will utilize DeFi lending protocols (eg. Compound, Aave) to lend the underlying assets of the SmartFunds and generate revenue from the interest accrued.

[7] (*Net Asset Value – NAV Definition*, n.d.)

Alvara's default position on receipt of underlying assets is that they will be held and not used for any staking, lending, yield enhancement or other function UNLESS the value of the token can be ensured and the liquidity of the token can be assured.

If yield enhancement opportunities arise and the value of the token can be protected via insurance, collateral or other means, and the liquidity at short notice can be maintained, such an opportunity can then be taken advantage of.

Any additional yield is always assessed in light of the token value and liquidity criteria.

The primary role of Alvara is the creation and management of SmartFunds and thus any endeavor which could undermine this function will not be entertained.

To summarize: if there are occasions where the integrity of the SmartFund is maintained and token value and liquidity are sufficiently addressed, such yield generation opportunities will be explored to enhance the returns on the SmartFund.

3.8 Alvara Analysis Tool

Alvara's SmartFund analysis tool is our proprietary research tool for backtesting and evaluating the performances of custom built SmartFunds. This is for use by Alvara fund managers and will be available to third party fund managers for a nominal subscription paid in the ALVA token.

Input: Mapping (tokens -> weights), backtest period, rebalance interval in days. Output: Performances Table - Return and Risk KPIs; Graphs - Tokens' returns correlation matrix, portfolio value over time, volatility comparison for a selected period.

3.9 SmartFund Liabilities and TER Calculation

Alvara's SmartFund liabilities can be broken down into the following categories:

Operational expenses:

- Management fees
- Trading fees
- Legal fees
- Government fees
- Accounting fees
- Auditor fees
- Software subscription fees
- Custodial service fees
- Marketing
- Development
- Security
- Employee/staff

SmartFund expenses:

Fees incurred during creation and redemption processes:

- Gas fees
- Transfer fees sending USDT from and to Alvara wallet
- DEX fees
- CEX fees
- Deposit/withdrawal fees
- Trading fees

Total Expense Ratio (TER) formula:

The total expense ratio (TER) describes a fund's operating costs relative to its assets.[8] It is a measure of a fund's operational efficiency.

Investors pay attention to the expense ratio to determine if a fund is an appropriate

investment for them after fees are considered.

TER is also known as the "net expense ratio" or "after reimbursement expense ratio.[9]

$$TER = \frac{Total\ Fund\ Costs}{Total\ Fund\ Assets}$$

3.10 Underlying Asset Selection

All Alvara Spot SmartFunds are 100% collateralized with the underlying assets included in the indexes that each SmartFund tracks. These underlying assets are purchased on a combination of both centralized and decentralized exchanges, connected to Alvara's protocol via APIs.

The eligible universe of tokens is filtered using two primary criteria:

- market cap
- liquidity (average daily traded volume ADTV)

A 50% weighting is applied to each of the primary criteria to ensure that they are given

[8] (Brown, n.d.)
[9] (Brown, 2021)

equal prominence in determining the universe.

There is no need to look at available supply in relation to total supply of token since liquidity can account for this.

The following secondary criteria will be applied if deemed necessary:

- recognised exchange listings
- publicly visible management teams

In this way we can ensure that the primary criteria are market cap and liquidity and "Eligible Market" could be divided into segments by market cap e.g. Large caps which are > $1bn, Mid-caps which are >$250m and small caps which are > $100m etc. (>$100m is currently top 400 tokens with about 85% of the market value).

Liquidity will exclude many of these. The ADTV for large caps: >$25m; mid caps: >$10m; small caps: >$5m.

The exchange selected for underlying asset purchase is determined at the time of execution based on both price and liquidity criteria.

Once purchased, underlying assets are held in Alvara's Fireblocks wallets, and employed in yield generation strategies in order to generate additional revenue for the funds, and dividends for SmartFund holders.

A "proof-of-reserve" integration with Chainlink ensures that balances of holding wallets are displayed publicly on the HiveX. This provides trustless proof of 100% underlying asset collateralization for all spot SmartFunds listed on the platform.

3.10.1 SmartFund Asset Potential

Through the construction of the HiveX framework, the potential SmartFund combinations are practically limitless. In fact the only limit is the speed at which real world assets are becoming

digitized. We fully anticipate the deployment of smart funds on the HiveX containing previously illiquid assets that have been made liquid through digitization. Examples include gold, real estate, infrastructure, bonds and carbon certificates, however the opportunities are limitless. Any digitized asset that can be purchased on any CEX or DEX that also meets Alvara's underlying asset inclusion criteria can be included in smart funds.

3.11 Rebalancing

Rebalancing of each SmartFund will take place monthly to the "strategic asset allocation". These are the starting weights for the SmartFund's underlying assets: e.g. 30% BTC, 30% ETH, 35% ADA, 5% ALVA. "Actual allocation" will vary during the month based on price moves.

At the end of the month the "eligible market" is recalculated to check that all the tokens still adhere to the market cap and liquidity criteria.

If assets remain eligible then rebalancing is simply the buying and selling of the existing underlying assets in the SmartFund to rebalance to the "Strategic Asset Allocation".

If during the month, the market cap or liquidity elements of a token change to result in them NOT qualifying for the "eligible market", the token will be flagged and an intra-month buffer of 20% will be applied to the token's market cap and liquidity requirements i.e. if market threshold was $100m and the tokens dips below that, then the token is flagged and monitored and a buffer is applied to it resulting in the threshold being $80m. This is to allow for extreme volatility intra-month of crypto markets. If the threshold plus buffer is breached for 5 consecutive days then the token is replaced with the next eligible token from the "Eligible Market" for an equivalent value.

The same 20% buffer will also apply to ADTV.

To summarize: rebalancing will happen monthly unless criteria intra-month are breached

after which there is a 20% buffer applied plus this intra-month buffer criteria in turn has to be breached for 5 consecutive days. Only if this happens will intra-month rebalancing occur. Intra-month rebalancing will be very rare and only carried out under exceptional circumstances. The 20% buffer level and 5 consecutive days can also be additional elements for the addendum and can also be different for different market segments i.e. there may be a very new market segment just starting and the market cap criteria could be very low and the buffer could be more and the consecutive days also more.

“Eligible Market” monitoring will be carried out by the Alvara team and using an oracle integration with Chainlink.

3.12 Leaderboard

Alvara’s leaderboard will rank all SmartFunds according to different metrics, allowing users to see each fund and fund manager’s track record.

Each fund will provide the following data within the leaderboard: underlying assets and their weightings; market cap; market price; share NAV; 24 hour volume; % change (1 hour, 24 hour, 7 days); cumulative return (1 month, 3 months, 6 months, YTD, Annualized); max drawdown (%).

3.13 Integrations

Chainlink:

1. Price Feeds
2. Verifiable Random Function (VRF)
3. Proof-of-Reserve

Fireblocks:

1. Vaults

2. DeFi API (Yield Generation)

3.14 Alvara APIs

Multiple purposes:

Market makers:
Allow Alvara and other fund managers to provide 3rd party market makers or liquidity providers with API to control funds in their HiveX account.

Traders:
Allow platform users to connect to third party software such as 3commas to automate trading strategies.

Description of functions:

- account: read account info
- trade: read order information; trade
- withdraw: read deposit/withdrawal information; withdraw
- transfer: read transfer information; transfer
- name: users must give each API key a unique name

3.15 Fiat Gateway Integration

An integrated fiat gateway allows users to easily invest in crypto funds directly from within the platform. Supplied by a third party provider, the fiat on ramp complies with all AML and CFT regulations within Alvara's jurisdiction.

3.16 NFT Integration

NFT integration provides users with access to lucrative revenue streams in the v2 platform. The NFT range consists of an indefinite number of futuristic artworks, depicting robotic creatures.

NFTs serve one main function with the protocol - proof-of-management (PoM) rights.

PoM NFTs:

- proof-of-management in v2 SmartFunds
- fund management rights transfer

3.16.1 Proof-of-Management (PoM NFTs)

Alvara's NFTs serve as a fund manager's proof-of-management. Upon the successful launch of a user-built SmartFund, the fund manager will receive an NFT connected to their fund. Managers must stake their NFT on the platform to gain access to both the management rights and the fee share. PoM NFTs are transferable and can be listed by users on Alvara's NFT Marketplace.

This transferability instills monetary value on both the management rights and the fee sharing structures that exist within the protocol.

3.17 V2 Fee Sharing Structure

V2 fund managers are incentivized to design, manage and promote their SmartFunds through the protocol's built-in fee sharing structure. Managers can earn up to a maximum

of 25% of the 0.1% trading fees, and the 0.5% annual management fee.

Market projections based on TradFi vs DeFi comparative analysis show that the top 100 funds in cryptocurrency's ETF industry can expect an average AUM of $254,300,094, with daily volumes of $12,374,507. If one of these funds were listed on the HiveX V2, then the fund manager could expect to earn a very lucrative $3965/day (nearly $1.5million per annum.)

3.18 Security

Leading cybersecurity consultancy firm, Hacken, are Alvara's appointed security auditors. They will carry out full security audits on all ecosystem smart contracts including token, bridge, staking and NFTs, as well as all on-chain contracts featured in the HiveX. The firm will also be carrying out full pentesting of the Protocol's website and exchange.

4. Alvara Core Behavior

4.1 Gamification

Alvara's advanced leaderboard provides the platform with a native ranking system, allowing fund managers to compete against one another for top ranking positions.

4.1.1 Leaderboards

The HiveX leaderboard is built into the platform and will showcase all listed SmartFunds with performance-based ranking.

4.1.2 Non-Fungible Tokens (NFTs)

The system of gamification is not limited to a leaderboard ranking by performance. NFT integration also plays a key role within the platform. As well as serving as proof-of-ownership, NFTs provide a fee sharing structure for holders. Incentives in the form of upgraded NFTs serve to reward fund managers as they achieve certain milestones. These upgrades allow the manager to earn a larger share of the transaction fees and develop a strong reputation as a seasoned portfolio creator.

The gamification of Alvara's HiveX is a crucial organic element to the ALVA Token's value. NFTs on the Alvara platform serve the following purposes:

- Proof-of-ownership of the management and fee sharing rights to a SmartFund
- Badges based on achievements and rankings

5. Legal Disclaimer

The information provided in this whitepaper and on the Alvara website does not constitute financial advice, trading advice, investment advice, legal advice, or any sort of advice whatsoever, and you should not treat any of the whitepaper or website content as such. The Alvara team does take every precaution to secure its token and protect investors and applies the same standards to vetting other projects that utilize its applications. Alvara does not endorse or recommend that any cryptocurrency be bought, sold, or held by you or any persons. Always do your own research and conduct your own due diligence to determine a project's success prospects. By purchasing ALVA, or any asset offered on the platform, you agree that you are not purchasing a security or investment. You also agree to hold the team harmless and not liable for any losses or taxes you may incur. You also agree that the team is presenting the token "as is" and is not legally required to provide any support or services. You

should have no expectation in any form, including guarantees of safety, price, functionality, or future feasibility of the ALVA token or any tokens launched on the Alvara platform. Always make sure that you comply with your local laws and regulations before you make any purchase.